

# **Experimental Setups & Usability Testing**

Detailed Notes on Human-Computer Interaction Research Methods

**Based on Lecture Slides: Techniques for Building in Usability**

Comprehensive Study Guide

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# 1 Introduction

In the field of Human-Computer Interaction (HCI), creating a usable product requires rigorous testing and research. This document outlines various experimental setups and techniques used to build usability into a product. These range from early-stage qualitative research to late-stage quantitative validation.

## 2 Techniques for Building in Usability

There are ten primary techniques identified for ensuring usability throughout the design lifecycle:

1. Ethnographic Research
2. Participatory Design
3. Focus Group Research
4. Surveys
5. Walkthroughs
6. Open and Closed Card Sorting
7. Paper Prototyping
8. Expert or Heuristic Evaluations
9. Usability Testing
10. Follow-up Studies

### 2.1 1. Ethnographic Research

Ethnographic research is a **qualitative method** derived from anthropology. It involves observing users in their natural environment ("in situ") rather than in a laboratory.

**Key Characteristics:**

- **Contextual Observation:** Observing users where they would actually use the product (e.g., an office, a home, a factory).
- **Data Gathering:** Focuses on the target user, their specific tasks, and their goals.
- **Deliverables:** The data collected is often used to create:
  - **Personas:** Fictional characters representing user types.
  - **Profiles:** Demographic data of users.
  - **Scenarios:** Stories describing how a user interacts with a system.
  - **Task Descriptions:** Detailed breakdowns of workflows.

### Real-World Example: Hospital Management System

Instead of asking doctors how they use software in a survey, a researcher stands in the emergency room (out of the way) and watches the doctor interact with the tablet while treating patients. They notice that the doctor struggles to tap small buttons while wearing latex gloves—a detail that might be missed in a lab.

## 2.2 2. Participatory Design

Participatory design (Co-design) brings the users into the design team. It creates a sense of ownership and ensures the product meets actual needs.

### Key Characteristics:

- Users are "on board" with the design team.
- Taps into the user's deep knowledge, skill sets, and emotional reactions.
- **Workshops:** Collaborative sessions where users, designers, and developers work together on specific aspects of the design.

## 2.3 3. Focus Group Research

Focus groups are used at the **early stages** of a project. They are primarily used to evaluate preliminary concepts and generate ideas.

### Key Characteristics:

- **Brainstorming:** Representative users discuss concepts.
- **Proof of Concept:** Used to validate if an idea has merit before coding begins.
- **Tools:** Low-fidelity tools are used, such as paper/pencil drawings, storyboards, or plastic models.

## 2.4 4. Surveys

Surveys are quantitative tools used at the **initial stage** of product implementation to gather data from a large population.

### Key Characteristics:

- Checks preferences of a broad base of users regarding an existing or potential product.
- **Question Types:**
  - *Open-ended:* Text fields allowing free responses.
  - *Closed-ended:* Multiple choice or Likert scales (e.g., 1-5 rating).
- Produces **Quantitative Data** (statistics, percentages).

## 2.5 5. Walkthroughs

Walkthroughs are often used at the **prototype stage**. This involves walking a user (or a team member) through a specific task flow.

### Key Characteristics:

- Envisions the user's route through an early concept.
- Used to develop work guides.
- Participants often assume specific roles (Role-playing) to test different perspectives.

## 2.6 6. Card Sorting

Card sorting is a technique used specifically to design the **Information Architecture (IA)** and improve "findability" of content.

### Types of Card Sorting:

- **Open Card Sorting:** Participants are given cards with content but **no** pre-existing categories. They must group the cards and *name the groups* themselves. This reveals the user's mental model.
- **Closed Card Sorting:** Participants are given cards and a set of **pre-existing categories**. They must sort the content into these fixed buckets. This tests if the terminology is clear.

### Real-World Example: E-Commerce Menu Design

An online store has 100 products.

- **Open Sort:** Users are given cards like "iPhone," "Charger," and "Headphones." One user creates a pile named "Mobile Accessories." Another names it "Gadgets." The designer uses the most common name.
- **Closed Sort:** Users are told to put "iPhone" into either "Electronics" or "Communication."

## 2.7 7. Paper Prototyping

A low-fidelity method where the interface is drawn on paper.

### Key Characteristics:

- Users are shown a paper interface and asked to perform tasks.
- **Wizard of Oz technique:** An expert (human computer) pulls up a new piece of graph paper representing the "next screen" based on where the user taps.

## 2.8 8. Expert or Heuristic Evaluations

A review of a product or system performed by an expert rather than a user.

### Key Characteristics:

- **Double Specialist:** ideally, the evaluator is an expert in both Usability (HCI principles) and the specific Domain (e.g., Medical software, Accounting).
- Often uses "Heuristics" (Rules of Thumb), such as Nielsen's 10 Usability Heuristics, to identify problems.

## 3 Usability Testing

Usability testing refers to techniques used to collect empirical data while observing end-users using the product.

### 3.1 Purpose and Goals

- **Confirm/Refute Hypothesis:** Using formal tests as true experiments.
- **Iterative Improvement:** Cycles of tests expose deficiencies, gradually shaping the product.
- **Informing Design:** providing data to back design decisions.
- **Eliminating Frustration:** Solving design problems before launch.
- **Improving Profitability:** A usable product sells better and requires less support.

### 3.2 Methodology: Classical vs. Practical

#### 3.2.1 The Classical Scientific Approach

Strict experimental setups require:

- Formulation of a specific hypothesis.
- Random assignment of participants to experimental conditions.
- Tight controls to eliminate confounding variables.
- Use of Control Groups.
- Sufficient sample size for statistical significance.

#### 3.2.2 Basic Elements of Practical Usability Testing

In industry, testing is often more flexible than academic experiments:

- **Research Questions:** Often replaces a formal hypothesis (e.g., "Can they find the button?").
- **Representative Sample:** Users who match the target demographic (may not be strictly random).
- **Environment:** Representation of the actual work environment.
- **Observation:** Watching users review or use the product representation.
- **Interviewing:** Extensive probing by the test moderator ("Why did you click that?").
- **Data Collection:** Both quantitative (time, errors) and qualitative (preferences, verbal comments).

### 3.3 Limitations of Usability Testing

- **Artificiality:** Testing is always an artificial situation; users may behave differently when watched (Hawthorne Effect).
- **No Guarantee of Success:** Results do not prove a product works, only that specific testers could use it.
- **Representation:** Participants are rarely fully representative of the entire target population.



## 4 Types of Usability Tests

Usability testing occurs in a cycle: *Prototyping* → *Design* → *Testing*. There are four specific types of tests, categorized by when they occur in the lifecycle and their objective.

### 4.1 1. Exploratory or Formative Study

**When:** Early in the development cycle.

**Objective:** To examine the effectiveness of the **preliminary design** or concept. It is "Formative" because it helps *form* the design.

**Focus Areas:**

- Does the interface support the user's goals?
- Does it communicate the intended workflow?
- Can the user navigate from screen to screen?
- **Mental Models:** How do users conceive the product?

**Methodology:**

- **Prototypes:** Horizontal representations (broad overview of features, shallow depth).
- **Walkthroughs:** Vertical representations (deep dive into one specific feature).
- **User-Oriented Questions:** "Is this 'walk up and use' or do I need a manual?"

### 4.2 2. Assessment or Summative Test

**When:** Early or midway through development (after fundamental design is established).

**Objective:** To expand on exploratory findings by evaluating lower-level operations. It focuses on *how well* a user can perform full-blown, realistic tasks.

**Methodology:**

- Users perform actual tasks rather than just commenting.
- **Reduced Interaction:** The moderator intervenes less than in exploratory tests to measure actual behavior.
- **Data:** Collection of quantitative measures (Time on task, error rates).

### 4.3 3. Validation or Verification Test

**When:** Late in the deployment cycle (near launch).

**Objective:** To measure usability against **established benchmarks**. It verifies that problems found earlier are fixed and no new ones were introduced.

**Methodology:**

- **Benchmarks:** Standards are identified prior to testing (e.g., "Login must take less than 10 seconds").
- **Zero Interaction:** The moderator acts as a silent observer.
- **Focus:** Strict collection of quantitative data to prove the system meets requirements.

## 4.4 4. Comparison Test

**When:** Can be performed at any point in the lifecycle (not specific).

**Objective:** Used to compare two or more designs (A/B Testing) or to compare against a competitor.

**Methodology:**

- Forces the design team to challenge conceptions.
- Participants consider *why* one design is better than the other.
- Can compare Interface Style A vs. Interface Style B, or Digital App vs. Paper Manual.

### Real-World Example: Comparison Test Example

A streaming service wants to change their video player.

- **Group A** uses the old player.
- **Group B** uses the new player with gesture controls.
- The team compares which group abandons videos less frequently.